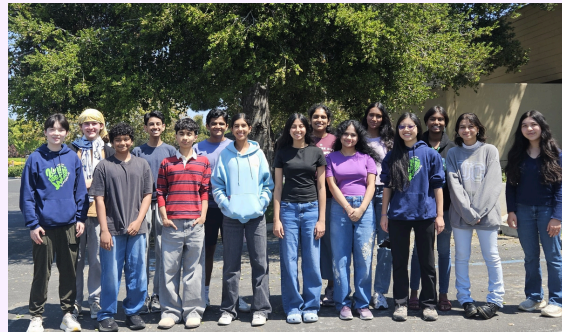


Environmental impacts

- Environmentally, the meat industry drives deforestation for grazing land, which in turn destroys habitats and reduces biodiversity
- Decomposing chicken releases methane (a greenhouse gas)
- Farms release harmful chemicals and waste products that pollute the atmosphere, waterways, and soil

Our Team

- We are the only high school iGEM team in Northern California
- Based at Khan Lab School in Mountain View, including students from nine high schools across the San Francisco Bay Area



The Cluck is Ticking

KhanLab-BayArea



English

The Problem

- Food waste costs the meat industry an estimated 94 billion USD every year
- Inaccurate expiration date labels often mislead consumers
- Date labels are typically conservative estimates, which can lead to the unnecessary disposal of safe meat or the accidental consumption of spoiled meat

Our Project

- Our project is to engineer a biosensor to detect spoilage in meat as an alternative to traditional expiration dates
 - The biosensor measures cadaverine, an organic compound produced as meat decomposes
 - As cadaverine levels rise, the biosensor responds by producing a blue chromoprotein, delivering a clear visual result
- In the future, we aim to adapt the biosensor for other meats

Consequences

- Consuming spoiled chicken can lead to infections from bacteria such as Salmonella and E. coli
- It is estimated that 1 in 9 people get sick after ingesting contaminated meat, with 1.52 million dying annually